

latinR proposal 2026: Flipping the simple features ‘plotting script’ with ggregions, Evangeline Reynolds

Status quo mapping with ggplot2’s `geom_sf()`

ggplot2 is a beloved graphical system because of its elegance and flexibility. Usually a graph requires the user to specify three elements (and ggplot2 can take care of the rest):

1. **data** - which dataframe will be the basis of the graphic
2. **aesthetic mapping** - should variables be represented by x position, y position, color, linewidth etc, and
3. **geom**, the mark that should take on these aesthetics.

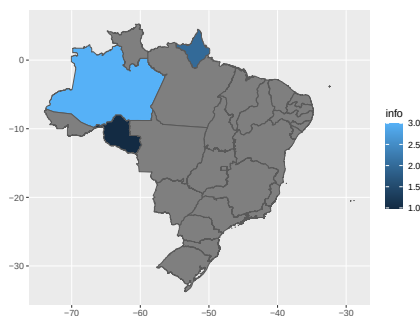
ggplot2 added more geographic viz capabilities with the addition of its `geom_sf_*()` functions. However, geographic mapping with `geom_sf` may feel a little out-of-step with ggplot2’s other plot builds. Let’s consider creating a choropleth of characteristic that varies with geographic area. Here’s example syntax.

```
library(tidyverse)

brazil_info_to_map <- tribble(~state, ~info,
  "Rondônia", 1,
  "Amapá", 2,
  "Amazônias", 3 )

## some geographic shape data
br_regiones <- geobr::read_state(year = 2020)

# get
brazil_info_to_map |>
  full_join(br_regiones,
    by = join_by(state == name_state)) |>
  ggplot() +
  aes(geometry = geom,
    fill = info) +
  geom_sf()
```



A first step is to find shape data for the areas of interest, ideally in a data frame with one column identifying the geographic area, and another list-column often named `geometry` with the geographic information included.

Then the actual data of interest is joined to the geography data frame via the identifying column. Then this data can be input into `ggplot()`. To render a map, `geom_sf()` can be added with no aesthetic mapping required, because under the hood, this layer is directed to look for a column named `geometry`. But to make the map a choropleth, aesthetic mapping instruction `aes(fill = my_var)` would be required.

Whereas so many geoms *require* at least one positional aesthetic (x and y) to be declared in plot syntax, `sf` weirdly doesn't - as it's all managed with precomputation and under-the-hood column discovery.

A new way: geography-specific layers (when aesthetic mapping is really about *mapping*)

This talk will show the `ggregions` workflow that's more similar to the 'classic' `ggplot2` build by requiring that the users define regions of interest. This package provides a new aesthetic, `region` and layer, `geom_region()`.

```
library(ggregions)

# declare relevant regions and their unique ids
br_regions |>
  select(state_name = name_state,
         state_abbr = abbrev_state,
         geometry = geom) |>
  set_regions()

ggplot(data = brazil_info_to_map) +
  stamp_region() +
  aes(region = state,
     fill = info) +
  geom_region()

last_plot() +
  stamp_region_border(keep = "Rondônia", color = "red") +
  stamp_region_label(aes(label = after_stat(state_abbr)))
```

